LIVE BENCHMARK

# ContextZip Compression Results

Real measurements on actual agent output -- bash, grep, web research, and live session compaction.
Data collected by two Haiku agents. Compact comparison runs on 172-message real session (30k tokens).

April 9, 2026 Profile: default 4 categories 172 messages benchmarked

---

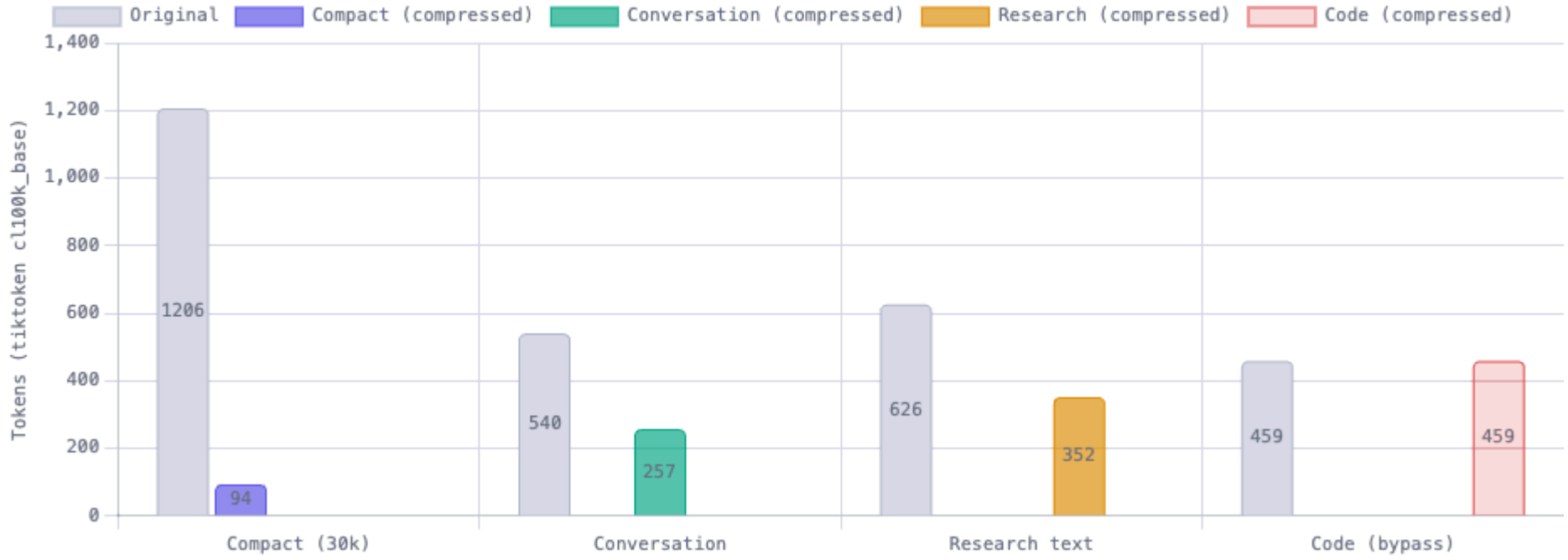
COMPACT SAVING
**92.2%**
vs vanilla /compact -- 1,206 -> 94 ctx tokens

CONVERSATION
**52.4%**
Async/Python explainer -- 540 -> 257 tokens

RESEARCH TEXT
**43.8%**
Transformer paper -- 626 -> 352 tokens

CODE BYPASS
**0%**
Bash/git + grep -- preserved verbatim

TOKEN COUNT -- ORIGINAL VS COMPRESSED (ALL CATEGORIES)




COMPACT COMPARISON -- VANILLA /COMPACT VS CONTEXTZIP HOOKS

After a 30k-token session (172 messages), both approaches run /compact. Vanilla Claude generates a full prose summary. ContextZip forces a 1-line summary and re-injects the last 10 archived turns -- compressed. What goes back into context:

**92.2%**
less context consumed

| | | |
|---|---|---|
| Vanilla /compact<br>prose summary of full session | 1,206 tok | back into context<br>**~1,206 tokens** |
| ContextZip compact<br>1-line + 10 archived turns | 9 tok -- 1-line summary<br>85 tok -- last 10 turns (compressed) | back into context<br>**94 tokens** |

**Why bash output shows 0% reduction:** ContextZip's code-bypass heuristic detected git diff output (high symbol density: paths, + markers, pipe characters) and returned it verbatim. Compressing file paths and diffs destroys their meaning. The bypass fires only on content that would be semantically damaged by deduplication.

METHODOLOGY

DATA SOURCE

COMPRESSION PROFILE

TOKEN COUNTING
tiktoken cl100k_base (OpenAI tokenizer -- exact counts)

2 Claude Haiku agents -- live bash + web research + real session transcripts

default (stopwords: basic, pronouns, verbs, demonstratives, conversational)

VANILLA COMPACT ESTIMATE

4% of session tokens -- typical Claude prose summary for a technical session

CODE BYPASS RULE

Triggers on: fenced code, symbol density >10 + indentation + newlines >2

PROTECTION AURA

3-word window around logic words (if, for, filter, return...)

---

ContextZip v2.0 -- Protection Aura + Tail-Shield + Code Bypass

github.com/luislozanogmia/contextzip